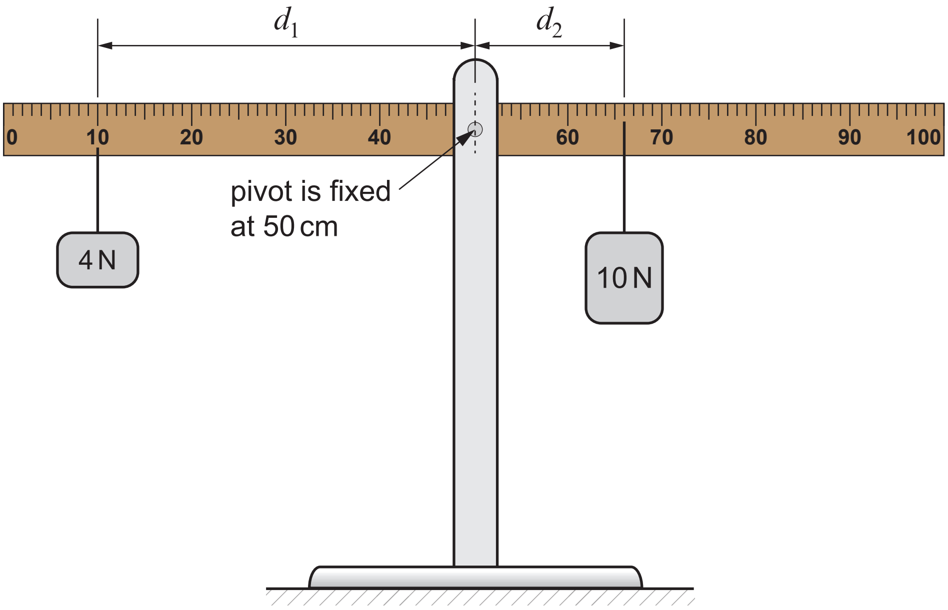
Examiner only

3. Mary investigated moments using a **100 cm** ruler and two different weights.

The apparatus used is shown below.



Mary checked the ruler was horizontal. She then attached weights to the ruler.
The 4 N weight, W_1, was attached to the ruler a distance, d_1, from the pivot.
The 10 N weight, W_2, was then attached to the ruler to make it horizontal again.
The distance, d_2, of the 10 N weight from the pivot was noted.

Mary's results are shown in the table.

Weight, W_1 (N)	Distance, d_1 (cm)	Weight, W_2 (N)	Distance, d_2 (cm)
4	40	10	16
4	35	10	14
4	20	10	8
4	15	10	6
4	5	10	2

(a) Mary states the resolution of the ruler she used in this experiment as 1 cm.
Use information from the diagram to explain whether Mary is correct. [1]

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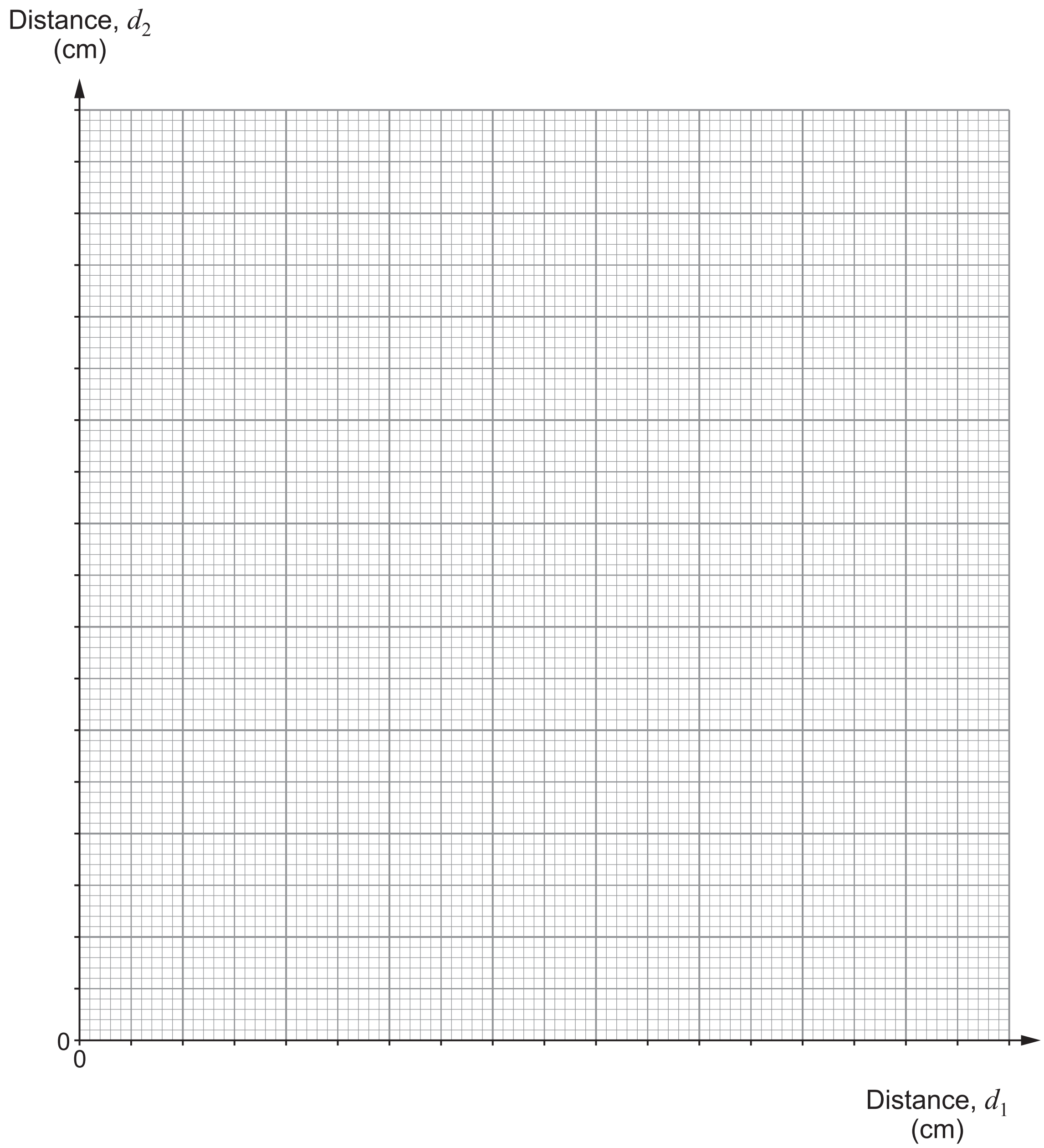
(b) (i) Plot the data on the grid below and draw a suitable **straight** line. [3]



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(ii) Mary suggests that the value of the gradient of the graph is the same as $\frac{W_1}{W_2}$.

Use data from the graph and the table to explain whether Mary is correct. [3]

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(c) (i) Mary now places the 10 N weight at a distance, d_2, of 32 cm from the pivot.
Use an equation from page 2 to calculate its clockwise moment about the pivot. [2]

Clockwise moment = ... N cm

(ii) Explain, using moments, why the ruler cannot now be balanced using the 4 N weight. [2]

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